

## System Monitoring and Performance Tuning

### Why System Monitoring is Essential?

System monitoring is essential for ensuring that a computer system operates efficiently and effectively. Regular monitoring helps in:

#### Early Detection of Issues:

By continuously tracking performance metrics, potential problems can be identified before they escalate into major failures or downtime.

#### Optimization of Resources:

Monitoring provides insights into resource utilization, allowing administrators to allocate resources more effectively, ensuring optimal system performance.

#### Capacity Planning:

Monitoring helps in understanding current resource usage and forecasting future needs, aiding in proactive planning for upgrades or scaling.

#### Improved Security:

Monitoring can help detect unauthorized access, potential breaches, or malicious activities, enhancing the overall security of the system.

#### Compliance and Reporting:

For organizations that need to comply with regulations, monitoring can help maintain logs and provide reports on system performance and security.

### CPU Usage

#### Definition:

This metric (measured) indicates the percentage of the CPU's processing power that is currently being utilized. High CPU usage may indicate that a system is under heavy load, while low usage may indicate underutilization.

## The University of Lahore

### Importance:

Monitoring CPU usage helps in identifying processes that are consuming excessive resources, allowing for performance tuning and troubleshooting.

### Commands for CPU Monitoring

- `top`
- `htop` (for a more user-friendly interface)
- `mpstat` (from the `sysstat` package)

### top Command

- The `top` command in Linux is a powerful tool used to monitor system processes and performance in real-time.
- It provides a dynamic view of your system's resource usage, such as CPU and memory consumption, and helps identify which processes are consuming the most resources.

### Detailed Breakdown of the top Command:

#### Basic Usage:

- To run the `top` command, simply open your terminal and type:  

```
top
```
- The screen will update in real time, with a summary of system usage at the top and a list of processes below.

#### System Summary Area:

The first five lines of the `top` output show an overview of the system:

#### Line 1: Uptime and Load Average

```
top - 10:15:20 up 2 days, 3:01, 2 users, load average: 0.44, 0.22, 0.15
```

**Time:** Current system time.

**Uptime:** How long the system has been running since the last reboot.

**Users:** Number of users logged into the system.

**Load Average:** The system load over the last 1, 5, and 15 minutes. Lower numbers indicate a lightly loaded system.

**The University of Lahore****Line 2: Task Summary**

Tasks: 234 total, 1 running, 230 sleeping, 0 stopped, 3 zombie

**Total:** The total number of processes.

**Running:** Number of processes currently running.

**Sleeping:** Number of processes that are idle.

**Stopped:** Number of stopped processes.

**Zombie:** Number of zombie (defunct) processes, which have completed execution but haven't been cleaned up by their parent process.

**Line 3: CPU Usage**

%Cpu(s): 2.3 us, 0.7 sy, 0.0 ni, 96.5 id, 0.3 wa, 0.0 hi, 0.2 si, 0.0 st

**us (User):** Time spent running non-kernel code (user space).

**sy (System):** Time spent running kernel code.

**ni (Nice):** Time spent running user processes at a "nice" priority level.

**id (Idle):** Time when the CPU is idle.

**wa (Wait):** Time spent waiting for I/O operations.

**hi (Hardware IRQ):** Time spent handling hardware interrupts.

**si (Software IRQ):** Time spent handling software interrupts.

**st (Steal):** Time stolen from the CPU by hypervisor for other tasks.

**Line 4: Memory Usage (RAM)**

MiB Mem : 16000.0 total, 8000.0 free, 3000.0 used, 5000.0 buff/cache

**Total:** Total available memory.

**Free:** Memory that is currently unused.

**Used:** Memory in use by processes.

**The University of Lahore**

**Buff/Cache:** Memory used by kernel buffers and cache.

**Line 5: Swap Usage**

MiB Swap: 4000.0 total, 2000.0 free, 1000.0 used, 1000.0 avail Mem

**Total:** Total amount of swap space.

**Free:** Unused swap space.

**Used:** Swap space in use.

**Avail Mem:** Available memory (includes both free memory and memory available for immediate use).

**Process List Area:**

The lower portion of the `top` output shows a list of running processes with the following columns:

**PID:** Process ID.

**USER:** Username of the process owner.

**PR (Priority):** Process priority.

**NI (Nice value):** Nice value (priority adjustment) of the process.

**VIRT:** Virtual memory used by the process.

**RES:** Resident memory used by the process (physical memory).

**SHR:** Shared memory used by the process.

**S (Status):** Process status:

**%CPU:** Percentage of CPU usage by the process.

**%MEM:** Percentage of RAM usage by the process.

**TIME+:** Total CPU time consumed by the process.

**COMMAND:** The command name or path of the process.

## The University of Lahore

### Interactive Commands in `top`:

While `top` is running, you can use several keyboard shortcuts to manipulate its output:

- h**: Help – Displays help and a list of all key commands.
- k**: Kill – Allows you to kill a process by typing its PID.
- u**: User – Filter processes by a specific user.
- P**: Sort by CPU usage (default).
- M**: Sort by memory usage.
- T**: Sort by the running time of processes.
- c**: Toggle between displaying the command name or the full command line.
- d**: Change the update delay interval.
- q**: Quite `top`.

### Customizing Output:

You can customize the columns displayed in the process list by pressing `f` or `o` and selecting/deselecting fields. You can also save your preferred layout by pressing `w`, which writes your configuration to `~/.toprc`.

### Running `top` with Options:

You can customize the behaviour of `top` using various command-line options:

- **-b (Batch Mode)**: Output is non-interactive, often used for scripting.

```
top -b -n 1
```

The `top -b -n 1` command is used to run the `top` command in **batch mode** for a **single iteration**. Let me break it down:

#### Explanation of `top -b -n 1`:

##### **-b (Batch Mode)**:

The `-b` option runs `top` in **batch mode**. In batch mode, `top` provides output in a non-interactive manner, suitable for logging or scripting.

Unlike the regular `top` command, which updates continuously and allows you to interact with the display (sorting, filtering, etc.), batch mode outputs the information once and exits.

**The University of Lahore****-n 1 (Number of Iterations):**

The `-n` option specifies the **number of iterations** (or updates) `top` should make before quitting.

`-n 1` means that `top` will only take a **single snapshot** of the system's processes and resource usage and then exit.

**Use Case:**

The command `top -b -n 1` is typically used for:

- **Logging system statistics:**

You can use this in scripts to capture the system's resource usage at a specific time.

**Non-interactive output:**

- Since batch mode does not require a terminal to stay open, you can redirect the output to a file or another command for further processing.

**Example:**

To save the output of `top` in batch mode to a file, you can run:

```
top -b -n 1 > top_output.txt
```

This command captures the system's resource usage and stores it in the file `top_output.txt`.

- **-n (Number of Iterations):** Limit the number of updates.

```
top -n 5
```

- **-u (User Filter):** Show processes for a specific user.

```
top -u username
```

**Common Use Cases for the `top` Command:****Identifying Resource-Heavy Processes:**

Use **`top`** to see which processes are consuming the most CPU or memory, helping you troubleshoot performance bottlenecks.

## The University of Lahore

## Monitoring System Health:

Constantly monitoring load averages, memory usage, and CPU utilization with `top` allows you to maintain system health and react to any issues proactively.

## Managing Processes:

You can directly kill processes from within `top` using the `k` command, which is useful when a process hangs or consumes excessive resources.

## Batch Mode for Scripting:

Use `top` in batch mode to capture snapshots of system processes at regular intervals. This is useful for creating logs or troubleshooting recurring performance issues.

## Practice Questions

- 1) How would you use the `top` command to find the process that is consuming the most CPU?

```

abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform: ~
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$ top

top - 21:38:33 up 8 min,  1 user,  load average: 0.68, 0.72, 0.49
Tasks: 340 total,   1 running, 339 sleeping,   0 stopped,   0 zombie
%Cpu(s):  3.4 us,  1.9 sy,  0.3 ni, 94.0 id,  0.2 wa,  0.0 hi,  0.3 si,  0.0 st
MiB Mem : 2852.7 total,  101.3 free, 1373.9 used, 1586.6 buff/cache
MiB Swap: 2852.0 total, 2852.0 free,   0.0 used, 1478.7 avail Mem

  PID USER      PR  NI   VIRT   RES   SHR  S  %CPU  %MEM     TIME+ COMMAND
 4074 root        20   0 1998252 47976 26200 S 17.2   1.6   0:10.31 snapd
2499 abdulla+   20   0 4392264 271128 134496 S  5.6   9.3   0:24.58 gnome-shell
2926 abdulla+  20   0 429412   30752 22072 S  3.0   1.1   0:02.67 snapd-desktop-i
3770 abdulla+  20   0 14524   6008  3756 R  1.3   0.2   0:01.41 top
4630 fwupd-r+   20   0 590896 29552 18116 S  1.3   1.0   0:00.78 fwupdmg
3478 abdulla+ 30  10 867812 247344 116464 S  1.0   8.5   1:59.15 update-manager
 336 root        20   0      0      0      0 S  0.7   0.0   0:01.25 jbd2/sda2-8
2706 abdulla+  20   0 216976 40896 30528 S  0.7   1.4   0:01.64 vmtotl
3768 _apt       25   5 25904 10720  9572 S  0.7   0.4   0:10.25 http
  15 root        20   0      0      0      0 I  0.3   0.0   0:01.24 rcu_preempt
  36 root        20   0      0      0      0 S  0.3   0.0   0:00.33 ksoftirqd/3
288 root        20   0      0      0      0 I  0.3   0.0   0:00.77 kworker/u514:28-writeback
 657 systemd+  20   0 17572  7804  6872 S  0.3   0.3   0:00.47 systemd-oomd
 741 root        20   0 244760  9516  8056 S  0.3   0.3   0:01.30 vmtotl
1015 message+   20   0 12224  7556  4784 S  0.3   0.3   0:02.60 dbus-daemon
3462 abdulla+  20   0 701860 60500 48372 S  0.3   2.1   0:03.57 gnome-terminal-
3748 root        25   5 224032 125424 81960 S  0.3   4.3   0:03.00 aptd
   1 root        20   0 23372 14836  9876 S  0.0   0.5   0:07.10 systemd
   2 root        20   0      0      0      0 S  0.0   0.0   0:00.05 kthreadd
   3 root        20   0      0      0      0 S  0.0   0.0   0:00.00 pool_workqueue_release
   4 root        0 -20      0      0      0 I  0.0   0.0   0:00.00 kworker/R-rcu_gp

```

- 2) Capture the current system processes and save them into a text file using the `top` command.

```

abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$ top -b -n 1 > processes.txt
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$

```

## The University of Lahore

3) How can you filter the top output to show only the processes owned by the user "john"?

```

abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform: ~
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$ top -u abdullah-sohail

top - 21:41:54 up 11 min,  1 user,  load average: 0.55, 0.60, 0.48
Tasks: 325 total,  1 running, 324 sleeping,  0 stopped,  0 zombie
%Cpu(s):  1.8 us,  1.0 sy,  0.4 ni, 96.6 id,  0.2 wa,  0.0 hi,  0.1 si,  0.0 st
MiB Mem : 2852.7 total,  98.6 free, 1354.1 used, 1610.7 buff/cache
MiB Swap: 2852.0 total, 2852.0 free,  0.0 used. 1498.5 avail Mem

  PID USER      PR  NI    VIRT    RES    SHR  S  %CPU  %MEM     TIME+ COMMAND
 2926 abdulla+  20   0  429412  30344  21664 S   1.0   1.0   0:05.65 snapd-desktop-i
 4910 abdulla+  20   0  14528   5984   3748 R   1.0   0.2   0:00.18 top
 3478 abdulla+  30  10  867812 247344 116464 S   0.7   8.5   2:01.48 update-manager
 2706 abdulla+  20   0  214448  38368  30528 S   0.3   1.3   0:02.39 vmtoolsd
 2899 abdulla+  20   0  389388   8272   7184 S   0.3   0.3   0:00.10 gvfs-afc-volume
 3462 abdulla+  20   0  701860  60560  48372 S   0.3   2.1   0:05.55 gnome-terminal-
 2267 abdulla+  20   0   21600  13128   9760 S   0.0   0.4   0:02.98 systemd
 2271 abdulla+  20   0   21476   3656   1848 S   0.0   0.1   0:00.00 (sd-pam)
 2280 abdulla+   9 -11 113604  12964   8716 S   0.0   0.4   0:00.18 pipewire
 2281 abdulla+  20   0   97744   6168   5284 S   0.0   0.2   0:00.05 pipewire
 2286 abdulla+  20   0   39140  11916  10408 S   0.0   0.4   0:02.43 snapd-desktop-i
 2290 abdulla+   9 -11 406816  19072  14056 S   0.0   0.7   0:00.35 wireplumber
 2292 abdulla+   9 -11 110520  12048   9496 S   0.0   0.4   0:00.08 pipewire-pulse
 2301 abdulla+  20   0   10856   6708   4780 S   0.0   0.2   0:01.61 dbus-daemon
 2302 abdulla+  20   0   316516 10504   9416 S   0.0   0.4   0:00.14 gnome-keyring-d
 2342 abdulla+  20   0   610300   7676   6888 S   0.0   0.3   0:00.14 xdg-document-po
 2348 abdulla+  20   0   309316   6380   5856 S   0.0   0.2   0:00.03 xdg-permission-
 2354 abdulla+  20   0   235676   6260   5748 S   0.0   0.2   0:00.02 gdm-wayland-ses
 2362 abdulla+  20   0   298244  16932  14996 S   0.0   0.6   0:00.09 gnome-session-b
 2439 abdulla+  20   0   162660   7016   6320 S   0.0   0.2   0:00.02 gcr-ssh-agent
 2440 abdulla+  20   0   91556   5800   5344 S   0.0   0.2   0:00.01 gnome-session-c

```

4) How can you change the refresh interval (update delay) of the top command to 5 seconds?

```

abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform: ~
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$ top -d 5

top - 21:44:00 up 13 min,  1 user,  load average: 0.49, 0.52, 0.46
Tasks: 322 total,  1 running, 321 sleeping,  0 stopped,  0 zombie
%Cpu(s):  4.1 us,  1.7 sy,  0.4 ni, 93.3 id,  0.4 wa,  0.0 hi,  0.2 si,  0.0 st
MiB Mem : 2852.7 total,  72.7 free, 1343.9 used, 1648.1 buff/cache
MiB Swap: 2852.0 total, 2852.0 free,  0.0 used. 1508.8 avail Mem

  PID USER      PR  NI    VIRT    RES    SHR  S  %CPU  %MEM     TIME+ COMMAND
 4074 root        20   0 2071984  48788  26200 S  17.9   1.7   0:59.68 snapd
 2499 abdulla+  20   0 4392264 271608 134496 S   2.4   9.3   0:34.83 gnome-shell
 2926 abdulla+  20   0 429412  30032  21352 S   1.8   1.0   0:07.71 snapd-desktop-i
  287 root        20   0      0      0      0 I   1.4   0.0   0:00.43 kworker/u514:27-events_unbound
 3462 abdulla+  20   0 701860  60560  48372 S   1.0   2.1   0:08.25 gnome-terminal-
 3478 abdulla+  30  10  867812 247344 116464 S   1.0   8.5   2:02.57 update-manager
  336 root        20   0      0      0      0 S   0.8   0.0   0:03.65 jbd2/sda2-8
 3768 _apt        25   5  25904  10720   9572 S   0.8   0.4   0:18.51 http
 5131 abdulla+  20   0 14524   5972   3744 R   0.8   0.2   0:00.16 top
  741 root        20   0 244760   9516   8056 S   0.4   0.3   0:02.16 vmtoolsd
 3748 root        25   5 224032 125424  81960 S   0.4   4.3   0:03.99 aptd
  121 root        20   0      0      0      0 I   0.2   0.0   0:00.79 kworker/3:2-mm_percpu_wq
  285 root        20   0      0      0      0 I   0.2   0.0   0:00.61 kworker/u514:25-flush-8:0
  418 root        20   0      0      0      0 I   0.2   0.0   0:00.75 kworker/2:2-mm_percpu_wq
  611 root        20   0      0      0      0 I   0.2   0.0   0:01.08 kworker/u513:4-ext4-rsv-conversion
  657 systemd+   20   0  17572   7804   6872 S   0.2   0.3   0:00.72 systemd-oomd
 1015 message+  20   0  12224   7556   4784 S   0.2   0.3   0:03.64 dbus-daemon
 2706 abdulla+  20   0 217024  40852  30528 S   0.2   1.4   0:02.80 vmtoolsd
 3793 root        0 -20      0      0      0 I   0.2   0.0   0:00.11 kworker/2:2H-kblockd
 3795 root        20   0      0      0      0 I   0.2   0.0   0:00.44 kworker/1:0-mm_percpu_wq
    1 root        20   0 23372  14840   9876 S   0.0   0.5   0:09.29 systemd
    2 root        20   0      0      0      0 S   0.0   0.0   0:00.05 kthreadd

```



## The University of Lahore

- 5) Using the top command, how would you kill a process with a specific PID, for example, PID 1234?

```

abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform: ~
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$ top

top - 21:45:56 up 15 min,  1 user,  load average: 0.15, 0.38, 0.42
Tasks: 324 total,  1 running, 323 sleeping,  0 stopped,  0 zombie
%Cpu(s):  2.0 us,  0.8 sy,  0.1 ni, 96.9 id,  0.2 wa,  0.0 hi,  0.1 si,  0.0 st
MiB Mem : 2852.7 total,  68.8 free, 1342.6 used, 1653.7 buff/cache
MiB Swap: 2852.0 total, 2852.0 free,  0.0 used, 1510.1 avail Mem
PID to signal/kill [default pid = 4074] 1234

  PID USER      PR  NI  VIRT  RES  SHR  S  %CPU  %MEM    TIME+  COMMAND
 4074 root        20   0 2071984 49100 26200 S   7.0   1.7   1:11.49 snapd
2499 abdulla+   20   0 4392252 271616 134496 S   1.3   9.3   0:37.38 gnome-shell
2926 abdulla+  20   0 429412 30032 21352 S   1.0   1.0   0:09.19 snapd-desktop-i
3462 abdulla+  20   0 702244 60824 48372 S   1.0   2.1   0:09.41 gnome-terminal-
3478 abdulla+  30  10 867812 247344 116464 S   1.0   8.5   2:03.85 update-manager
 336 root        20   0      0      0      0 S   0.7   0.0   0:04.34 jbd2/sda2-8
3768 _apt        25   5 25904 10720 9572 S   0.7   0.4   0:19.21 http
 121 root        20   0      0      0      0 I   0.3   0.0   0:00.86 kworker/3:2-mm_percpu_wq
 741 root        20   0 244760 9516 8056 S   0.3   0.3   0:02.38 vmtoolsd
1015 message+  20   0 12224 7556 4784 S   0.3   0.3   0:03.89 dbus-daemon
2706 abdulla+  20   0 217176 41004 30528 S   0.3   1.4   0:03.13 vmtoolsd
3748 root        25   5 224032 125424 81960 S   0.3   4.3   0:04.28 aptd
5156 abdulla+  20   0 14524 5984 3760 R   0.3   0.2   0:00.04 top
   1 root        20   0 23372 14840 9876 S   0.0   0.5   0:09.33 systemd
   2 root        20   0      0      0      0 S   0.0   0.0   0:00.07 kthreadd
   3 root        20   0      0      0      0 S   0.0   0.0   0:00.00 pool_workqueue_release
   4 root        0 -20      0      0      0 I   0.0   0.0   0:00.00 kworker/R-rcu_gp
   5 root        0 -20      0      0      0 I   0.0   0.0   0:00.00 kworker/R-sync_wq
   6 root        0 -20      0      0      0 I   0.0   0.0   0:00.00 kworker/R-kvfree_rcu_reclaim
   7 root        0 -20      0      0      0 I   0.0   0.0   0:00.00 kworker/R-slub_flushwq
   8 root        0 -20      0      0      0 I   0.0   0.0   0:00.00 kworker/R-netns
   9 root        20   0      0      0      0 I   0.0   0.0   0:00.68 kworker/0:0-cgroup_free

```

```

abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform: ~
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$ top

top - 21:47:44 up 17 min,  1 user,  load average: 0.25, 0.35, 0.40
Tasks: 323 total,  1 running, 322 sleeping,  0 stopped,  0 zombie
%Cpu(s):  3.8 us,  1.6 sy,  0.4 ni, 93.7 id,  0.4 wa,  0.0 hi,  0.1 si,  0.0 st
MiB Mem : 2852.7 total,  73.2 free, 1341.8 used, 1650.3 buff/cache
MiB Swap: 2852.0 total, 2852.0 free,  0.0 used, 1510.8 avail Mem

  PID USER      PR  NI  VIRT  RES  SHR  S  %CPU  %MEM    TIME+  COMMAND
 4074 root        20   0 2071984 48560 26200 S  15.8   1.7   1:24.89 snapd
2499 abdulla+   20   0 4392300 271508 134496 S   3.0   9.3   0:40.71 gnome-shell
2926 abdulla+  20   0 429412 30032 21352 S   2.3   1.0   0:10.90 snapd-desktop-i
3462 abdulla+  20   0 702244 60824 48372 S   1.3   2.1   0:10.87 gnome-terminal-
 336 root        20   0      0      0      0 S   1.0   0.0   0:05.15 jbd2/sda2-8
3478 abdulla+  30  10 867812 247344 116464 S   1.0   8.5   2:04.85 update-manager
 741 root        20   0 244760 9516 8056 S   0.7   0.3   0:02.66 vmtoolsd
2706 abdulla+  20   0 217176 41004 30528 S   0.7   1.4   0:03.52 vmtoolsd
3768 _apt        25   5 25904 10720 9572 S   0.7   0.4   0:20.46 http
5156 abdulla+  20   0 14524 5988 3760 R   0.7   0.2   0:00.39 top
   30 root        20   0      0      0      0 S   0.0   0.0   0:00.10 ksoftirqd/2
 435 root       -51   0      0      0      0 S   0.3   0.0   0:00.44 irq/16-vmwgfx
 651 root        20   0      0      0      0 I   0.3   0.0   0:00.80 kworker/u513:8-writeback
1015 message+  20   0 12224 7556 4784 S   0.3   0.3   0:04.18 dbus-daemon
2958 abdulla+  20   0 261020 89148 61920 S   0.3   3.1   0:01.05 Xwayland
3748 root        25   5 224032 125424 81960 S   0.3   4.3   0:04.70 aptd
3795 root        20   0      0      0      0 I   0.3   0.0   0:00.62 kworker/1:0-mpt_poll_0
   1 root        20   0 23372 14840 9876 S   0.0   0.5   0:09.34 systemd
   2 root        20   0      0      0      0 S   0.0   0.0   0:00.07 kthreadd
   3 root        20   0      0      0      0 S   0.0   0.0   0:00.00 pool_workqueue_release
   4 root        0 -20      0      0      0 I   0.0   0.0   0:00.00 kworker/R-rcu_gp
   5 root        0 -20      0      0      0 I   0.0   0.0   0:00.00 kworker/R-sync_wq

```

- 6) How can you customize the top command to display the full command line (including arguments) for each process instead of just the command name?

## The University of Lahore

```

abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform: ~
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$ top
top - 21:48:51 up 18 min, 1 user, load average: 0.40, 0.38, 0.41
Tasks: 319 total, 1 running, 318 sleeping, 0 stopped, 0 zombie
%Cpu(s): 3.4 us, 1.6 sy, 0.3 ni, 94.2 id, 0.4 wa, 0.0 hi, 0.1 si, 0.0 st
MiB Mem : 2852.7 total, 82.5 free, 1347.8 used, 1634.9 buff/cache
MiB Swap: 2852.0 total, 2852.0 free, 0.0 used, 1504.8 avail Mem

  PID USER      PR  NI   VIRT   RES   SHR  S  %CPU  %MEM    TIME+  COMMAND
 4074 root        20   0 2071984 48876 26200 S 12.2   1.7   1:35.29 /usr/lib/snapd/snapd
2499 abdulla+   20   0 4392316 271612 134496 S 3.9   9.3   0:44.79 /usr/bin/gnome-shell
2926 abdulla+  20   0 429412 30032 21352 S 1.6   1.0   0:12.24 /snap/snapd-desktop-integration/343/usr/bin/snapd-desktop-integration
 336 root        20   0      0      0      0 S 1.0   0.0   0:05.81 [jbd2/sda2-8]
3478 abdulla+  30  10 867812 247344 116464 S 1.0   8.5   2:06.45 /usr/bin/python3 /usr/bin/update-manager --no-update --no-focus-on-map
5158 abdulla+  20   0 14524 5996 3760 R 1.0   0.2   0:00.26 top
3462 abdulla+  20   0 702244 60888 48372 S 0.7   2.1   0:12.51 /usr/libexec/gnome-terminal-server
3748 root       25   5 224032 125424 81960 S 0.7   4.3   0:05.00 /usr/bin/python3 /usr/sbin/aptd
  15 root        20   0      0      0      0 I 0.3   0.0   0:01.83 [rcu_preempt]
287 root        20   0      0      0      0 I 0.3   0.0   0:00.50 [kworker/u514:27-events_freezable_pwr_efficient]
611 root        20   0      0      0      0 I 0.3   0.0   0:01.19 [kworker/u513:4-events_power_efficient]
657 systemd+   20   0 17572 7804 6872 S 0.3   0.3   0:00.94 /usr/lib/systemd/systemd-oomd
1015 message+   20   0 12224 7556 4784 S 0.3   0.3   0:04.38 @dbus-daemon --system --address=systemd: --nofork --nopidfile --systemd-
2706 abdulla+  20   0 222104 45932 30528 S 0.3   1.6   0:03.95 /usr/bin/vmtoolsd -n vmusr --blockFd 3 --uinputFd 4
3768 _apt        25   5 25904 10720 9572 S 0.3   0.4   0:21.14 /usr/lib/apt/methods/http
3795 root       20   0      0      0      0 I 0.3   0.0   0:00.70 [kworker/1:0-events]
4652 root       20   0      0      0      0 I 0.3   0.0   0:00.26 [kworker/0:2-events]
   1 root       20   0 23372 14840 9876 S 0.0   0.5   0:09.34 /sbin/init splash
   2 root       20   0      0      0      0 S 0.0   0.0   0:00.07 [kthreadd]
   3 root       20   0      0      0      0 S 0.0   0.0   0:00.00 [pool_workqueue_release]
   4 root       0 -20      0      0      0 I 0.0   0.0   0:00.00 [kworker/R-rcu_gp]

```

7) How can you sort the process list by memory usage instead of CPU usage in top?

```

abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform: ~
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$ top
top - 21:50:07 up 20 min, 1 user, load average: 0.35, 0.37, 0.40
Tasks: 319 total, 1 running, 318 sleeping, 0 stopped, 0 zombie
%Cpu(s): 3.7 us, 1.9 sy, 0.3 ni, 93.7 id, 0.3 wa, 0.0 hi, 0.1 si, 0.0 st
MiB Mem : 2852.7 total, 110.2 free, 1358.6 used, 1596.8 buff/cache
MiB Swap: 2852.0 total, 2852.0 free, 0.0 used, 1494.1 avail Mem

  PID USER      PR  NI   VIRT   RES   SHR  S  %CPU  %MEM    TIME+  COMMAND
2499 abdulla+   20   0 4392316 271636 134496 S 5.9   9.3   0:48.58 gnome-shell
3478 abdulla+  30  10 867812 247344 116464 S 1.0   8.5   2:07.50 update-manager
3748 root       25   5 224032 125424 81960 S 0.7   4.3   0:05.36 aptd
2958 abdulla+  20   0 261020 89156 61920 S 0.0   3.1   0:01.07 Xwayland
3275 abdulla+  20   0 723520 88508 70300 S 0.0   3.0   0:00.45 mutter-x11-fram
3229 abdulla+  20   0 643048 82132 65888 S 0.0   2.8   0:00.48 gsd-xsettings
2675 abdulla+  20   0 900068 64592 50908 S 0.0   2.2   0:01.09 evolution-alarm
3462 abdulla+  20   0 702244 60888 48372 S 1.0   2.1   0:13.69 gnome-terminal-
3201 abdulla+  20   0 3136460 58288 43620 S 0.0   2.0   0:01.86 gjs
4074 root       20   0 2071984 48812 26200 S 13.9   1.7   1:46.35 snapd
4637 root       20   0 614836 44688 37836 S 0.0   1.5   0:03.06 fwupd
2607 abdulla+  20   0 595360 43328 35192 S 0.0   1.5   0:01.39 evolution-sourc
3117 abdulla+  20   0 778452 41692 31272 S 0.0   1.4   0:00.72 xdg-desktop-por
2706 abdulla+  20   0 214540 38496 30528 S 0.3   1.3   0:04.31 vmtoolsd
3356 abdulla+  20   0 568740 32644 25000 S 0.0   1.1   0:00.44 update-notifier
2967 abdulla+  20   0 825472 30864 26788 S 0.0   1.1   0:00.47 evolution-addre
2808 abdulla+  20   0 421536 30392 19064 S 0.0   1.0   0:09.86 ibus-extension-
2926 abdulla+  20   0 429412 30032 21352 S 2.0   1.0   0:13.76 snapd-desktop-i
3115 abdulla+  39  19 736372 29968 18812 S 0.0   1.0   0:01.12 tracker-miner-f
3455 abdulla+  20   0 458940 29688 22320 S 0.0   1.0   0:00.37 gnome-terminal.
3158 abdulla+  20   0 2531512 27468 22712 S 0.0   0.9   0:00.18 gic

```

8) How can you use the top command to continuously monitor processes in batch mode for 3 iterations and save the output to a file?

```

abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$ top -b -n 3 > output.txt
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$

```

## htop Command

- **htop** is an interactive system-monitoring tool for Unix-like operating systems, including Linux.

**The University of Lahore**

- It provides a real-time, dynamic view of system processes and resource usage in a visually appealing and user-friendly interface compared to the standard `top` command.

**Key Features of *htop*:****Color-Coded Information:**

*htop* uses colours to represent different system metrics (measured) such as CPU usage, memory, and swap usage, which makes it easier to understand compared to the monochrome *top* utility.

**Graphical Representation:**

It includes bars to graphically represent the CPU, memory, and swap usage, which is a significant improvement over the *top* command's numeric representation.

**Real-Time Interaction:**

You can interact with *htop* using your keyboard to scroll vertically through the list of processes or horizontally to view full command lines.

**Easy Sorting:**

Processes can be sorted based on various criteria such as CPU usage, memory usage, process ID (PID), and more. You can change the sorting order with a single keypress.

**Killing Processes:**

*htop* makes it simple to kill processes without having to use a separate `kill` command. Just select the process and press the `F9` key to send a kill signal.

**Tree View of Processes:**

It offers a tree view of processes, where you can see the hierarchical structure of parent-child relationships between processes.

**Filtering and Searching:**

*htop* includes search and filtering features, allowing users to find processes by name, PID, or other attributes quickly. This is particularly useful on systems with a large number of processes.

## The University of Lahore

### Process Details:

For each process, **htop** shows detailed information such as the user running the process, the process's priority (nice value), its memory and CPU usage, and the command line used to start the process.

### Customizable Display:

You can customize which metrics are displayed and how they are organized, making it more flexible than **top**.

### Running **htop**

#### Installation:

**htop** may not be pre-installed on some Linux distributions, but it can be easily installed through the package manager:

#### On Debian/Ubuntu:

```
sudo apt-get install htop
```

#### On RHEL/CentOS:

```
sudo yum install htop
```

#### On Fedora:

```
sudo dnf install htop
```

**Starting htop:** To start **htop**, simply type **htop** in the terminal and hit enter:

```
htop
```

### Interface Layout

#### CPU and Memory Usage:

At the top of the screen, **htop** displays the system's current CPU and memory usage. It uses bars that are color-coded:

- **Blue:** Low-priority processes.
- **Green:** Normal processes.
- **Red:** Kernel or system-level processes.

**The University of Lahore****Task Information:**

Just below the CPU and memory usage bars, `htop` shows the number of tasks (processes), threads, load average, and uptime.

**Process List:**

The main section displays all running processes, including their **PID**, **user**, **CPU%**, **MEM%**, **time running**, and **command**.

**Footer:**

The footer of the `htop` interface includes function key shortcuts for quick actions:

- F1:** Help (displays help menu).
- F2:** Setup (customizes the interface).
- F3:** Search (find specific processes).
- F4:** Filter (filters displayed processes).
- F5:** Tree (toggles tree view for processes).
- F6:** Sort (sorts processes by selected criteria like CPU or memory usage).
- F7/F8:** Nice (adjusts process priority).
- F9:** Kill (sends a kill signal to a process).
- F10:** Quit (exits `htop`).

**Sorting Processes**

You can change how processes are sorted by pressing `F6` and selecting a criterion such as:

- **PERCENT\_CPU:** Sort by CPU usage.
- **PERCENT\_MEM:** Sort by memory usage.
- **TIME+:** Sort by the amount of CPU time the process has consumed.
- **PID:** Sort by process ID.

## The University of Lahore

**Practice Questions**

1) What is htop, and how does it differ from the top command?

| Feature         | top           | htop                  |
|-----------------|---------------|-----------------------|
| Interface       | Simple, plain | Colorful, interactive |
| Mouse support   | ✗ No          | ✓ Yes                 |
| Scrolling       | ✗ No          | ✓ Yes                 |
| Process killing | Manual        | Easy (function keys)  |
| Tree view       | ✗ No          | ✓ Yes                 |

2) What color represents low-priority processes in htop?

**Blue color** → represents **low-priority (nice) processes**

3) How can you install htop on a Debian-based system?

```
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$ sudo apt install htop
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Suggested packages:
  lm-sensors
The following NEW packages will be installed:
  htop
0 upgraded, 1 newly installed, 0 to remove and 223 not upgraded.
Need to get 171 kB of archives.
After this operation, 434 kB of additional disk space will be used.
Get:1 http://pk.archive.ubuntu.com/ubuntu noble/main amd64 htop amd64 3.3.0-4build1 [171 kB]
Fetched 171 kB in 3s (58.2 kB/s)
Selecting previously unselected package htop.
(Reading database ... 158550 files and directories currently installed.)
Preparing to unpack ../htop_3.3.0-4build1_amd64.deb ...
Unpacking htop (3.3.0-4build1) ...
Setting up htop (3.3.0-4build1) ...
Processing triggers for desktop-file-utils (0.27-2build1) ...
Processing triggers for hicolor-icon-theme (0.17-2) ...
Processing triggers for gnome-menus (3.36.0-1ubuntu3) ...
Processing triggers for man-db (2.12.0-4build2) ...
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$
```

4) What key do you press to quit htop?

Press Q

## The University of Lahore

5) Describe the steps to start htop after installation.

```

abdullah-sohail@abdullah-sohail-Virtual-Platform: ~
0% Tasks: 118, 386 thr, 226 kthr; 1 running
1% Load average: 0.40 1.25 1.16
2% Uptime: 01:00:31
3%
Mem[ ] 827M/2.79G
Swap[ ] 295M/2.79G

Main 170
PID USER      PRI  NI  VIRT   RES   SHR  S  CPU% MEM%   TIME+  Command
8451 abdullah-s 20    0 11756 5544 3700 R  8.9  0.2  0:01.30 htop
2499 abdullah-s 20    0 4291M 127M 64812 S  0.0  4.5  4:17.58 /usr/bin/gnome-shell
7902 abdullah-s 20    0 686M 61208 48608 S  0.6  2.1  0:03.62 /usr/libexec/gnome-terminal-server
2534 abdullah-s -21   0 4291M 127M 64812 S  0.6  4.5  0:21.43 /usr/bin/gnome-shell
2547 abdullah-s 20    0 4291M 127M 64812 S  0.0  4.5  0:04.45 /usr/bin/gnome-shell
2706 abdullah-s 20    0 214M 25380 18184 S  0.6  0.9  0:12.01 /usr/bin/vmtoolsd -n vmusr --blockFd 3 --uiinputFd 4
1 root        20    0 23520 14084 9852 S  0.0  0.5  0:13.83 /sbin/init splash
386 root      19   -1 67656 14620 1896 S  0.0  0.5  0:02.19 /usr/lib/systemd/systemd-journald
425 root      20    0 148M 1708 1424 S  0.0  0.1  0:00.00 vmware-vmblock-fuse /run/vmblock-fuse -o rw,subtype=vmware-vmblock,default_permissions,allow_other,dev,suid
426 root      20    0 148M 1708 1424 S  0.0  0.1  0:00.00 vmware-vmblock-fuse /run/vmblock-fuse -o rw,subtype=vmware-vmblock,default_permissions,allow_other,dev,suid
427 root      20    0 148M 1708 1424 S  0.0  0.1  0:00.00 vmware-vmblock-fuse /run/vmblock-fuse -o rw,subtype=vmware-vmblock,default_permissions,allow_other,dev,suid
447 root      20    0 31620 8052 4936 S  0.0  0.3  0:01.54 /usr/lib/systemd/systemd-udevd
657 systemd-oo 20    0 17572 7296 6848 S  0.0  0.2  0:03.36 /usr/lib/systemd/systemd-oond
659 systemd-re 20    0 21852 13348 10900 S  0.0  0.5  0:00.94 /usr/lib/systemd/systemd-resolved
660 systemd-tl 20    0 91060 7740 7076 S  0.0  0.3  0:00.19 /usr/lib/systemd/systemd-timesyncd
713 systemd-tl 20    0 91060 7740 7076 S  0.0  0.3  0:00.00 /usr/lib/systemd/systemd-timesyncd
734 root      20    0 56076 11476 9908 S  0.0  0.4  0:00.29 /usr/bin/VGAAuthService
741 root      20    0 239M 8960 7500 S  0.6  0.3  0:10.48 /usr/bin/vmtoolsd

```

6) How do you sort processes by memory usage in htop?

Inside htop: F6

```

abdullah-sohail@abdullah-sohail-Virtual-Platform: ~
0% Tasks: 118, 386 thr, 200 kthr; 1 running
1% Load average: 0.04 0.62 0.92
2% Uptime: 01:04:13
3%
Mem[ ] 829M/2.79G
Swap[ ] 292M/2.79G

Main 170
Sort by PID
PID USER      PRI  NI  VIRT   RES   SHR  S  CPU% MEM%   TIME+  Command
8451 abdullah-s 20    0 11756 5616 3764 R  7.3  0.2  0:20.18 htop
2706 abdullah-s 20    0 214M 26468 18184 S  0.6  0.9  0:12.74 /usr/bin/vmtoolsd -n vmusr --blockFd 3 --uiinputFd 4
4640 root        20    0 600M 25512 21100 S  0.6  0.9  0:00.20 /usr/libexec/fwupd/fwupd
7902 abdullah-s 20    0 687M 61244 48608 S  0.6  2.1  0:05.92 /usr/libexec/gnome-terminal-server
NICE 7902 abdullah-s 20    0 687M 61244 48608 S  0.6  2.1  0:05.92 /usr/libexec/gnome-terminal-server
M_VIRT 1 root        20    0 23520 14084 9852 S  0.0  0.5  0:13.87 /sbin/init splash
M_RESIDENT 386 root      19   -1 67656 14780 14056 S  0.0  0.5  0:02.20 /usr/lib/systemd/systemd-journald
M_SHARE 425 root      20    0 148M 1708 1424 S  0.0  0.1  0:00.00 vmware-vmblock-fuse /run/vmblock-fuse -o rw,subtype=vmware-vmblock,default_permissions,allow_other,dev,suid
STATE 426 root      20    0 148M 1708 1424 S  0.0  0.1  0:00.00 vmware-vmblock-fuse /run/vmblock-fuse -o rw,subtype=vmware-vmblock,default_permissions,allow_other,dev,suid
PERCENT_CPU 427 root      20    0 148M 1708 1424 S  0.0  0.1  0:00.00 vmware-vmblock-fuse /run/vmblock-fuse -o rw,subtype=vmware-vmblock,default_permissions,allow_other,dev,suid
PERCENT_MEM 447 root      20    0 31620 8052 4936 S  0.0  0.3  0:01.54 /usr/lib/systemd/systemd-udevd
TIME 657 systemd-oo 20    0 17572 7296 6848 S  0.0  0.2  0:03.58 /usr/lib/systemd/systemd-oond
Command 659 systemd-re 20    0 21852 13348 10900 S  0.0  0.5  0:00.94 /usr/lib/systemd/systemd-resolved
660 systemd-tl 20    0 91060 7740 7076 S  0.0  0.3  0:00.19 /usr/lib/systemd/systemd-timesyncd
713 systemd-tl 20    0 91060 7740 7076 S  0.0  0.3  0:00.00 /usr/lib/systemd/systemd-timesyncd
734 root      20    0 56076 11476 9908 S  0.0  0.4  0:00.29 /usr/bin/VGAAuthService
741 root      20    0 239M 8960 7500 S  0.0  0.3  0:11.07 /usr/bin/vmtoolsd
818 root      20    0 239M 8960 7500 S  0.0  0.3  0:00.00 /usr/bin/vmtoolsd
915 root      20    0 239M 8960 7500 S  0.0  0.3  0:00.36 /usr/bin/vmtoolsd
916 root      20    0 239M 8960 7500 S  0.0  0.3  0:00.00 /usr/bin/vmtoolsd
1010 avahi      20    0 8676 4172 3940 S  0.0  0.1  0:00.19 avahi-daemon: running [abdullah-sohail-Virtual-Platform.local]
Enter/Sort Esc/Cancel

```

7) What is the purpose of the F9 key in htop?

F9 = Kill process



## The University of Lahore

8) How can you filter processes in htop?

Press inside htop: F4

The screenshot shows the htop interface with the 'Filter' menu (F4) open. The filter is set to 'ab', which filters the process list to show only processes owned by the user 'abdullah'. The process list includes the 'avahi-daemon' and three instances of 'ibus-daemon'.

| PID  | USER       | PRI | NI | VIRT | RES  | SHR  | S | CPU% | MEM% | TIME+   | Command                                                               |
|------|------------|-----|----|------|------|------|---|------|------|---------|-----------------------------------------------------------------------|
| 1010 | avahi      | 20  | 0  | 867K | 4172 | 3940 | S | 0.0  | 0.1  | 0:00.19 | avahi-daemon: running [abdullah-sohail-VMware-Virtual-Platform.local] |
| 2621 | abdullah-s | 20  | 0  | 379M | 7856 | 6996 | S | 0.0  | 0.3  | 0:01.51 | /usr/bin/ibus-daemon --panel disable                                  |
| 2677 | abdullah-s | 20  | 0  | 379M | 7856 | 6996 | S | 0.0  | 0.3  | 0:00.00 | /usr/bin/ibus-daemon --panel disable                                  |
| 2678 | abdullah-s | 20  | 0  | 379M | 7856 | 6996 | S | 0.0  | 0.3  | 0:00.00 | /usr/bin/ibus-daemon --panel disable                                  |
| 2681 | abdullah-s | 20  | 0  | 379M | 7856 | 6996 | S | 0.0  | 0.3  | 0:02.36 | /usr/bin/ibus-daemon --panel disable                                  |

9) Explain how you can view the process tree in htop.

Press inside htop: F5

The screenshot shows the htop interface with the 'List' menu (F5) open. The process list is now displayed as a tree structure, showing the hierarchy of processes. The root process is 'init', which has children like 'VGAuthService', 'python3', and 'vmttoolsd'. The 'python3' process has children for 'unattended-upgrades' and 'snapd'. The 'vmttoolsd' process has children for 'polkitd' and 'snapd'.

| PID  | USER    | PRI | NI | VIRT  | RES   | SHR  | S | CPU% | MEM% | TIME+   | Command                                                                                       |
|------|---------|-----|----|-------|-------|------|---|------|------|---------|-----------------------------------------------------------------------------------------------|
| 1    | root    | 20  | 0  | 2352K | 14084 | 9852 | S | 0.0  | 0.5  | 0:13.90 | /sbin/init splash                                                                             |
| 734  | root    | 20  | 0  | 5607K | 11476 | 9908 | S | 0.0  | 0.4  | 0:00.29 | /usr/bin/VGAuthService                                                                        |
| 1610 | root    | 20  | 0  | 115M  | 13248 | 9460 | S | 0.0  | 0.5  | 0:00.28 | /usr/bin/python3 /usr/share/unattended-upgrades/unattended-upgrade-shutdown --wait-for-signal |
| 1650 | root    | 20  | 0  | 115M  | 13248 | 9460 | S | 0.0  | 0.5  | 0:00.00 | /usr/bin/python3 /usr/share/unattended-upgrades/unattended-upgrade-shutdown --wait-for-signal |
| 741  | root    | 20  | 0  | 239M  | 8960  | 7500 | S | 0.0  | 0.3  | 0:11.96 | /usr/bin/vmttoolsd                                                                            |
| 818  | root    | 20  | 0  | 239M  | 8960  | 7500 | S | 0.0  | 0.3  | 0:00.00 | /usr/bin/vmttoolsd                                                                            |
| 915  | root    | 20  | 0  | 239M  | 8960  | 7500 | S | 0.0  | 0.3  | 0:00.39 | /usr/bin/vmttoolsd                                                                            |
| 916  | root    | 20  | 0  | 239M  | 8960  | 7500 | S | 0.0  | 0.3  | 0:00.00 | /usr/bin/vmttoolsd                                                                            |
| 1066 | polkitd | 20  | 0  | 381M  | 12128 | 8296 | S | 0.0  | 0.4  | 0:01.26 | /usr/lib/polkit-1/polkitd --no-debug                                                          |
| 1335 | polkitd | 20  | 0  | 381M  | 12128 | 8296 | S | 0.0  | 0.4  | 0:00.19 | /usr/lib/polkit-1/polkitd --no-debug                                                          |
| 1336 | polkitd | 20  | 0  | 381M  | 12128 | 8296 | S | 0.0  | 0.4  | 0:00.00 | /usr/lib/polkit-1/polkitd --no-debug                                                          |
| 1339 | polkitd | 20  | 0  | 381M  | 12128 | 8296 | S | 0.0  | 0.4  | 0:00.31 | /usr/lib/polkit-1/polkitd --no-debug                                                          |
| 4074 | root    | 20  | 0  | 2023M | 22580 | 9384 | S | 0.0  | 0.8  | 0:02.30 | /usr/lib/snapd/snapd                                                                          |
| 4083 | root    | 20  | 0  | 2023M | 22580 | 9384 | S | 0.0  | 0.8  | 0:41.07 | /usr/lib/snapd/snapd                                                                          |
| 4084 | root    | 20  | 0  | 2023M | 22580 | 9384 | S | 0.0  | 0.8  | 0:57.94 | /usr/lib/snapd/snapd                                                                          |
| 4085 | root    | 20  | 0  | 2023M | 22580 | 9384 | S | 0.0  | 0.8  | 1:00.87 | /usr/lib/snapd/snapd                                                                          |
| 4086 | root    | 20  | 0  | 2023M | 22580 | 9384 | S | 0.0  | 0.8  | 0:00.00 | /usr/lib/snapd/snapd                                                                          |
| 4087 | root    | 20  | 0  | 2023M | 22580 | 9384 | S | 0.0  | 0.8  | 0:00.00 | /usr/lib/snapd/snapd                                                                          |
| 4088 | root    | 20  | 0  | 2023M | 22580 | 9384 | S | 0.0  | 0.8  | 0:55.80 | /usr/lib/snapd/snapd                                                                          |
| 4096 | root    | 20  | 0  | 2023M | 22580 | 9384 | S | 0.0  | 0.8  | 0:05.23 | /usr/lib/snapd/snapd                                                                          |

10) What command would you use to find and kill a process with a specific name using htop?

F3 to find it and F9 to kill it



abduallah-sohail@abduallah-sohail-VMware-Virtual-Platform: ~

```

0[|||||] 5.7% Tasks: 116, 382 thr, 202 kthr; 1 running
1[|||||] 0.0% Load average: 0.11 0.17 0.54
2[|||||] 0.0% Uptime: 01:12:58
3[|||||] 1.3%
Mem[|||||] 823M/2.79G
Swap[|||||] 258M/2.79G

Main 170
PID USER PRI NI VIRT RES SHR S CPU% MEM% TIME+ Command
3171 abduallah-s 20 0 2608M 13872 13856 S 0.0 0.5 0:00.00 /usr/bin/gjs -n /usr/share/gnome-shell/org.gnome.ScreenSaver
2612 abduallah-s 20 0 2728M 13772 13752 S 0.0 0.5 0:00.32 /usr/bin/gjs -n /usr/share/gnome-shell/org.gnome.Shell.Notifications
2630 abduallah-s 20 0 2728M 13772 13752 S 0.0 0.5 0:00.00 /usr/bin/gjs -n /usr/share/gnome-shell/org.gnome.Shell.Notifications
2632 abduallah-s 20 0 2728M 13772 13752 S 0.0 0.5 0:00.00 /usr/bin/gjs -n /usr/share/gnome-shell/org.gnome.Shell.Notifications
2635 abduallah-s 20 0 2728M 13772 13752 S 0.0 0.5 0:00.00 /usr/bin/gjs -n /usr/share/gnome-shell/org.gnome.Shell.Notifications
2654 abduallah-s 20 0 2728M 13772 13752 S 0.0 0.5 0:00.00 /usr/bin/gjs -n /usr/share/gnome-shell/org.gnome.Shell.Notifications
2655 abduallah-s 20 0 2728M 13772 13752 S 0.0 0.5 0:00.00 /usr/bin/gjs -n /usr/share/gnome-shell/org.gnome.Shell.Notifications
2656 abduallah-s 20 0 2728M 13772 13752 S 0.0 0.5 0:00.00 /usr/bin/gjs -n /usr/share/gnome-shell/org.gnome.Shell.Notifications
2658 abduallah-s 20 0 2728M 13772 13752 S 0.0 0.5 0:00.00 /usr/bin/gjs -n /usr/share/gnome-shell/org.gnome.Shell.Notifications
2659 abduallah-s 20 0 2728M 13772 13752 S 0.0 0.5 0:00.00 /usr/bin/gjs -n /usr/share/gnome-shell/org.gnome.Shell.Notifications
2662 abduallah-s 20 0 2728M 13772 13752 S 0.0 0.5 0:00.00 /usr/bin/gjs -n /usr/share/gnome-shell/org.gnome.Shell.Notifications
2663 abduallah-s 20 0 2728M 13772 13752 S 0.0 0.5 0:00.00 /usr/bin/gjs -n /usr/share/gnome-shell/org.gnome.Shell.Notifications
2664 abduallah-s 20 0 2728M 13772 13752 S 0.0 0.5 0:00.00 /usr/bin/gjs -n /usr/share/gnome-shell/org.gnome.Shell.Notifications
2392 abduallah-s 20 0 309M 9700 8812 S 0.0 0.3 0:00.09 /usr/bin/gnome-keyring-daemon --foreground --components=pkcs11,secrets --control-directory=/run/user/10
2398 abduallah-s 20 0 309M 9700 8812 S 0.0 0.3 0:00.00 /usr/bin/gnome-keyring-daemon --foreground --components=pkcs11,secrets --control-directory=/run/user/10
2399 abduallah-s 20 0 309M 9700 8812 S 0.0 0.3 0:00.00 /usr/bin/gnome-keyring-daemon --foreground --components=pkcs11,secrets --control-directory=/run/user/10
2310 abduallah-s 20 0 309M 9700 8812 S 0.0 0.3 0:00.10 /usr/bin/gnome-keyring-daemon --foreground --components=pkcs11,secrets --control-directory=/run/user/10
2316 abduallah-s 20 0 309M 9700 8812 S 0.0 0.3 0:00.00 /usr/bin/gnome-keyring-daemon --foreground --components=pkcs11,secrets --control-directory=/run/user/10
2499 abduallah-s 20 0 4291M 129M 65000 S 1.3 4.5 4:32.08 /usr/bin/gnome-shell

```

11) How can you customize which metrics are displayed in htop?

abduallah-sohail@abduallah-sohail-VMware-Virtual-Platform: ~

```

0[|||||] 0.6% Tasks: 116, 382 thr, 201 kthr; 1 running
1[|||||] 5.7% Load average: 0.11 0.17 0.52
2[|||||] 0.0% Uptime: 01:13:59
3[|||||] 0.0%
Mem[|||||] 823M/2.79G
Swap[|||||] 258M/2.79G

Setup
Categories
Display options For current screen tab: Main
Header layout [X] Tree view
Meters [ ] - Tree view is always sorted by PID (htop 2 behavior)
Screens [ ] - Tree view is collapsed by default
Colors Global options:
[X] Show tabs for screens
[ ] Shadow other users' processes
[X] Hide kernel threads
[ ] Hide userland process threads
[ ] Hide processes running in containers
[X] Display threads in a different color
[ ] Show custom thread names
[X] Show program path
[ ] Highlight program "basename"
[X] Highlight out-dated/removed programs (red) / libraries (yellow)
[ ] Shadow distribution path prefixes
[ ] Merge exe, comm and cmdline in Command
[X] - Try to find comm in cmdline (when Command is merged)
[X] - Try to strip exe from cmdline (when Command is merged)
[X] Highlight large numbers in memory counters
F1 F2 F3 F4 F5 F6 F7 F8 F9 F10 Done

```

12) What information does htop display about each process? List at least four details.

htop shows many details, for example:

- **PID** → Process ID
- **USER** → Owner of process
- **CPU%** → CPU usage
- **MEM%** → Memory usage
- **TIME** → CPU time used
- **COMMAND** → Process name
- **PR (Priority)**
- **NI (Nice value)**

**The University of Lahore****mpstat Command**

- The `mpstat` command is a useful tool in Linux and Unix-like operating systems for monitoring CPU usage.
- It is part of the `sysstat` package and provides detailed statistics about CPU usage, including information on individual CPU cores as well as the overall CPU activity.

Here's a detailed overview of the ***mpstat*** command:

**Basic Syntax**

```
mpstat [options] [interval] [count]
```

- **options:** Various options to customize the output.
- **interval:** The time interval in seconds between reports (e.g., 1 for every second).
- **count:** The number of reports to generate (e.g., 5 for five reports).

**Key Features****CPU Utilization:**

Displays the percentage of CPU utilization for each individual processor as well as the overall average.

**Multi-Core Awareness:**

Capable of displaying statistics for each core separately, which is particularly useful for multi-core systems.

**Idle and I/O Wait Time:**

Shows the amount of time the CPU spends idle, waiting for I/O operations to complete.

**User and System Time:**

Breaks down CPU time into user processes, system processes, and other categories.

**Common Options**

- u:** Display CPU usage (this is the default).
- P ALL:** Show statistics for all CPUs.
- h:** Display output in a human-readable format.

**The University of Lahore**

**-v:** Display version information.

### Example Usage

#### Basic Usage:

```
mpstat
```

This command provides a one-time report of CPU usage for all processors.

#### Monitoring Every Second:

```
mpstat 1 5
```

This command provides CPU statistics every second for a total of five reports.

#### Showing All CPUs:

```
mpstat -P ALL 1
```

This command shows detailed statistics for each CPU core every second.

### Sample Output

When you run **mpstat**, you might see output similar to the following:

```
Linux 5.4.0-42-generic (hostname)      10/22/2024      _x86_64_      (4
CPU)
10:23:50 PM  CPU      %usr      %nice      %sys %iowait      %irq      %soft      %steal
%guest  %gnice  %idle
10:23:50 PM  all       12.50      0.00      8.75      0.50      0.00      0.25      0.00
0.00      0.00      77.00
10:23:50 PM    0       15.00      0.00      8.00      1.00      0.00      0.00      0.00
0.00      0.00      76.00
10:23:50 PM    1       10.00      0.00      9.00      0.00      0.00      0.50      0.00
0.00      0.00      80.50
10:23:50 PM    2       12.00      0.00      7.00      0.00      0.00      0.00      0.00
0.00      0.00      81.00
10:23:50 PM    3       14.00      0.00      8.00      0.00      0.00      0.00      0.00
0.00      0.00      78.00
```

### Interpretation of Columns

**%usr:** Percentage of CPU time spent on user processes.

**%nice:** Percentage of CPU time spent on user processes with a positive nice value.

## The University of Lahore

**%sys:** Percentage of CPU time spent on kernel processes.

**%iowait:** Percentage of time the CPU is idle while waiting for I/O operations to complete.

**%irq:** Percentage of CPU time spent handling hardware interrupts.

**%soft:** Percentage of CPU time spent handling software interrupts.

**%steal:** Percentage of time the virtual CPU waits for a real CPU while the hypervisor is servicing another virtual processor.

**%idle:** Percentage of time the CPU is idle.

## Practice Questions

- 1) If you notice that the %iowait value is consistently high (e.g., above 10%) in your mpstat output, what does that indicate, and what actions could you take?

ANSWER:

```

abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform: ~
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$ iostat
Linux 6.17.0-14-generic (abdullah-sohail-VMware-Virtual-Platform)      03/25/2026      _x86_64_      (4 CPU)

avg-cpu:  %user   %nice %system %iowait  %steal   %idle
            5.58    0.90    4.79    0.41    0.00   88.32

Device            tps    kB_read/s    kB_wrtn/s    kB_dscd/s    kB_read    kB_wrtn    kB_dscd
loop0              0.00         0.00         0.00         0.00         21         0         0
loop1              0.17         3.07         0.00         0.00        14244         0         0
loop10             0.77         4.55         0.00         0.00        21119         0         0
loop11             0.09         1.12         0.00         0.00         5183         0         0
loop12             0.18         6.55         0.00         0.00        30401         0         0
loop13             0.00         0.01         0.00         0.00         58         0         0
loop14             0.01         0.14         0.00         0.00         643         0         0
loop15             0.74        30.23         0.00         0.00       140275         0         0
loop16             0.01         0.09         0.00         0.00         397         0         0
loop17             0.01         0.24         0.00         0.00         1113         0         0
loop2              0.01         0.24         0.00         0.00         1130         0         0
loop3              0.02         0.25         0.00         0.00         1172         0         0
loop4              0.01         0.09         0.00         0.00         439         0         0
loop5              0.95        51.39         0.00         0.00       238456         0         0
loop6              0.02         0.11         0.00         0.00         498         0         0
loop7              0.01         0.24         0.00         0.00         1106         0         0
loop8              1.53        44.02         0.00         0.00       204225         0         0
loop9              0.01         0.08         0.00         0.00         382         0         0
sda                34.24       1026.71       1305.77         0.00      4763585      6058356         0
sr0                 0.01         0.45         0.00         0.00         2100         0         0
sr1                 0.01         0.45         0.00         0.00         2098         0         0

```

A high %iowait value indicates that the CPU is waiting for disk I/O operations, meaning the system is I/O-bound. This can be caused by slow storage or heavy disk usage. Solutions include optimizing disk usage, upgrading to faster storage like SSD, and increasing RAM.

## The University of Lahore

- 2) Write the command to monitor CPU statistics every 2 seconds for a total of 3 reports, including statistics for all CPUs.

```

abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform: ~
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$ mpstat -P ALL 2 3
Linux 6.17.0-14-generic (abdullah-sohail-VMware-Virtual-Platform)      03/25/2026      _x86_64_      (4 CPU)

10:49:05 PM CPU      %usr   %nice    %sys %iowait    %irq   %soft  %steal  %guest  %gnice   %idle
10:49:07 PM all      0.25    0.00    0.25   0.00    0.00   0.00   0.00   0.00   0.00   99.50
10:49:07 PM 0        0.51    0.00    1.02   0.00    0.00   0.00   0.00   0.00   0.00   98.48
10:49:07 PM 1        0.50    0.00    0.00   0.00    0.00   0.00   0.00   0.00   0.00   99.50
10:49:07 PM 2        0.00    0.00    0.00   0.00    0.00   0.00   0.00   0.00   0.00  100.00
10:49:07 PM 3        0.00    0.00    0.00   0.00    0.00   0.00   0.00   0.00   0.00  100.00

10:49:09 PM CPU      %usr   %nice    %sys %iowait    %irq   %soft  %steal  %guest  %gnice   %idle
10:49:09 PM all      0.51    0.00    1.27   0.00    0.00   0.00   0.00   0.00   0.00   98.22
10:49:09 PM 0        1.01    0.00    1.01   0.00    0.00   0.00   0.00   0.00   0.00   97.98
10:49:09 PM 1        1.02    0.00    1.53   0.00    0.00   0.00   0.00   0.00   0.00   97.45
10:49:09 PM 2        0.00    0.00    1.55   0.00    0.00   0.00   0.00   0.00   0.00   98.45
10:49:09 PM 3        0.00    0.00    1.02   0.00    0.00   0.00   0.00   0.00   0.00   98.98

10:49:09 PM CPU      %usr   %nice    %sys %iowait    %irq   %soft  %steal  %guest  %gnice   %idle
10:49:11 PM all      1.13    0.00    0.25   0.00    0.00   0.00   0.00   0.00   0.00   98.62
10:49:11 PM 0        3.03    0.00    0.51   0.00    0.00   0.00   0.00   0.00   0.00   96.46
10:49:11 PM 1        1.01    0.00    0.00   0.00    0.00   0.00   0.00   0.00   0.00   98.99
10:49:11 PM 2        0.00    0.00    0.00   0.00    0.00   0.00   0.00   0.00   0.00  100.00
10:49:11 PM 3        0.50    0.00    0.50   0.00    0.00   0.00   0.00   0.00   0.00   99.00

Average:   CPU      %usr   %nice    %sys %iowait    %irq   %soft  %steal  %guest  %gnice   %idle
Average:   all      0.63    0.00    0.59   0.00    0.00   0.00   0.00   0.00   0.00   98.78
Average:   0        1.52    0.00    0.84   0.00    0.00   0.00   0.00   0.00   0.00   97.64
Average:   1        0.84    0.00    0.51   0.00    0.00   0.00   0.00   0.00   0.00   98.65
Average:   2        0.00    0.00    0.51   0.00    0.00   0.00   0.00   0.00   0.00   99.49
Average:   3        0.17    0.00    0.50   0.00    0.00   0.00   0.00   0.00   0.00   99.33
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$

```

- 3) Explain the meaning of the %steal column in the mpstat output. Why might this value be significant?

ANSWRER:

%steal represents the percentage of CPU time taken by the hypervisor from a virtual machine. It indicates how much CPU time is unavailable to the VM due to other virtual machines. A high value suggests CPU contention and can lead to performance issues.